CELESTE SPACEWORKS — MISSION SCREENING REPORT
Vehicle: Lunar-capable demo (illustrative)
Generated: 2026-10-30 23:57:42 UTC
Selected analysis target: Low Earth Orbit

This is an educational engineering-screening report, not flight certification.

MISSION STATUS
Trajectory has not been run; this is a preflight screening report.
Selected mission target: Low Earth Orbit
Approximate target delta-v budget: 9.40 km/s
Estimated vehicle delta-v: 12.04 km/s
Initial thrust-to-weight: 4.819
Power margin: +2.05 kW

RANKED DESIGN ADVISOR
1. [HIGH] Ideal delta-v budget is short
   Evidence: Available 12.04 km/s; screening requirement 12.60 km/s; shortfall 0.56 km/s.
   Suggested action: Increase usable compatible propellant or engine Isp, then recheck wet mass and liftoff TWR.
2. [MEDIUM] Landing-simulation inputs are incomplete
   Evidence: Lunar-capable demo (illustrative): no landing gear / contact envelope defined; Lunar-capable demo (illustrative): allowable attitude (tilt) and angular rate not fully specified; Demo heavy engine: exit area or sea-level thrust unknown; cannot correct for ambient pressure; and 23 more blocking definition gap(s)
   Suggested action: Supply the missing landing-gear, attitude-control, and engine-envelope data before making powered-descent or touchdown claims.
3. [MEDIUM] Vehicle center of mass is unknown
   Evidence: At least one component with nonzero mass lacks center-of-mass data.
   Suggested action: Enter component center-of-mass values to enable aggregate balance and stability checks.
4. [LOW] Vehicle inputs are not fully traceable
   Evidence: 14 vehicle or component definition(s) use illustrative or estimated data.
   Suggested action: Replace showcase values with sourced data and record source revision and uncertainty where available.

Rule-based screening advice only; it is not flight certification or a real-vehicle design prescription.